Prescribing a power spectrum by choosing Fourier-mode variances. Let

$$u(x) = \sum_{n \in \mathbb{Z}^d} \widehat{u}_n e^{in\cdot x}$$

be a random velocity field, and assume isotropy in the sense that

$$\mathbb{E}\,|\widehat{u}_n|^2 = \sigma(|n|)^2.$$

Define the shell energy spectrum by

$$E(k) := \sum_{k \le |n| < k+1} \mathbb{E}\,|\widehat{u}_n|^2.$$

In the continuum this corresponds to

$$E(k) = \frac{d}{dk} \int_{|\xi| \le k} \mathbb{E}\,|\widehat{u}(\xi)|^2 \, d\xi = \int_{|\xi| = k} \mathbb{E}\,|\widehat{u}(\xi)|^2 \, dS(\xi).$$

Since a shell of radius k in dimension d contains $\sim k^{d-1}$ modes,

$$E(k) \sim k^{d-1} \sigma(k)^2.$$

Therefore, to impose a target power law

$$E(k) \sim k^{-p},$$

one should choose

$$\sigma(k)^2 \sim k^{-(p+d-1)}.$$

Equivalently,

$$\mathbb{E}\,|\widehat{u}_n|^2 \sim |n|^{-(p+d-1)}.$$

Relation with Hölder regularity. If one parametrizes the spectrum by a Hölder exponent α via

$$p = 2\alpha + 1,$$

then the natural variance choice is

$$\mathbb{E}\,|\widehat{u}_n|^2 \sim |n|^{-2\alpha-d},$$

and this gives

$$E(k) \sim k^{d-1} k^{-2\alpha-d} = k^{-2\alpha-1}.$$

Hence the spectral slope $-p$ and the Hölder exponent α are related by

$$p = 2\alpha + 1, \qquad \alpha = \frac{p-1}{2}.$$

Kolmogorov example. For the Kolmogorov law

$$E(k) \sim k^{-5/3},$$

one has

$$p = \frac{5}{3}, \qquad \alpha = \frac{1}{3}.$$

Thus in any dimension d one should set

$$\mathbb{E}\,|\widehat{u}_n|^2 \sim |n|^{-(d+2/3)},$$

which indeed yields

$$E(k) \sim k^{-5/3}.$$